

The Closure of the Ocean Mixed Layer Temperature Budget Using Level-Coordinate Model Fields

SEUNG-BUM KIM, ICHIRO FUKUMORI, AND TONG LEE

Jet Propulsion Laboratory, California Institute of Technology, Pasadena, California

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ABSTRACT

Entrainment is an important element of the mixed layer mass, heat, and temperature budgets. Conventional procedures to estimate entrainment heat advection often do not permit the closure of heat and temperature budgets because of inaccuracies in its formulation. In this study a rigorous approach to evaluate the effect of entrainment using the output of a general circulation model (GCM) that does not have an explicit prognostic mixed layer model is described. The integral elements of the evaluation are 1) the rigorous estimates of the temperature difference between mixed layer water and entrained water at each horizontal grid point, 2) the formulation of the temperature difference such that the budget closes over a volume greater than one horizontal grid point, and 3) the apparent warming of the mixed layer during the mixed layer shoaling to account for the weak vertical temperature gradient within the mixed layer.

This evaluation of entrainment heat advection is compared with the estimates by other commonly used ad hoc formulations by applying them in three regions: the north-central Pacific, the Kuroshio Extension, and the Niño-3 areas in the tropical Pacific. In all three areas the imbalance in the mixed layer temperature budget by the ad hoc estimates is significant, reaching a maximum of about 4 K yr^{-1} .

1. Introduction

The oceanic surface mixed layer has a relatively weak vertical stratification and is often characterized by large vertical mixing. The temperature averaged over the mixed layer (MLT) has been used widely as a proxy for sea surface temperature (SST). One of the reasons of studying MLT instead of SST is to avoid the effects of the very large vertical mixing within the mixed layer. Mixing within the mixed layer, being internal to the layer, redistributes surface heating within the layer but does not change the net heat content of the mixed layer. The mixed layer as a whole is dynamically more interesting as it interacts with the atmosphere (e.g., Zebiak and Cane 1987) and with the interior of the ocean (e.g., Deser et al. 2003). The budget of MLT is the subject of this study.

The MLT in various regions has been studied extensively by analyzing the components of the MLT tendency budget such as surface heat flux, advection, mix-

ing, and entrainment (Behringer and Stommel 1981; Qiu and Kelly 1993; Kessler et al. 1998; Alexander et al. 2000; Wang and McPhaden 2001; Kim et al. 2004). A prerequisite of a complete budget analysis is the closure of the budget, meaning that the sum of the budget components exactly matches the MLT tendency. However, such a prerequisite is often not fulfilled. For instance, Qiu (2002) found that the budget computed from observations did not close, and suggested that unresolved eddy flux and errors in the estimates may explain the budget imbalance. Moreover, analyses of the MLT budget are often incomplete. For example, Wang and McPhaden (2001) with observations and Vialard and Delecluse (1998) with a general circulation model (GCM) estimated mixing and/or entrainment as residuals of the MLT balance rather than estimating them explicitly. However, the residual term may contain errors in other components of the balance thus complicating the analysis of the MLT balance. In this study we focus on explicitly evaluating all the terms in the MLT balance using GCM products.

As numerical models satisfy first principles (e.g., closed budgets), any imbalance in the MLT budget indicates an incomplete formulation of the budget components. Among the budget components, our particular

Corresponding author address: Dr. S. B. Kim, Remote Sensing Systems, 438 First Street, Suite 200, Santa Rosa, CA 95401.
E-mail: kim@remss.com

interest in this study is the formulation of the entrainment heat advection (i.e., the heat energy required to convert cold thermocline water into warm mixed layer water; e.g., Kraus and Turner 1967). Entrainment is an important mechanism controlling the upper-ocean temperature. In midlatitudes, entrainment is typically large due primarily to the significant annual variation in the mixed layer volume. Entrainment compensates for 35% of annual heating of the ocean in the northwest Pacific (Qiu and Kelly 1993). Interannually entrainment explains the reemergence of the previous year's temperature signal (Deser et al. 2003) and a substantial fraction of the interannual warming in the north-central Pacific during the 1997–2000 period (Kim et al. 2004). In the tropical Pacific, Wang and McPhaden (2001) concluded that the residual of the MLT budget, considered to be entrainment and mixing tendencies (although these two were not separated from each other), was one of the largest terms in determining interannual variations of MLT in the Tropics.

Entrainment heat advection can be evaluated as the velocity of entraining water times the difference in temperature between mixed layer water and entraining water (ΔT). The evaluation of ΔT has been difficult mainly because the temperature of the entraining thermocline water is not known precisely. Consequently, ad hoc schemes have been used for ΔT , or an empirical relationship is constructed to infer the entrainment temperature (e.g., Swenson and Hansen 1999; Zhang et al. 2004). However, these ad hoc schemes often introduce errors, as noted by Swenson and Hansen (1999) and Qiu (2002). Thorough investigation is needed to address the issue of how well one can close the MLT budget.

In this study, we develop a rigorous scheme for evaluating the entrainment heat advection that allows the closure of the MLT budget exactly. The schemes are developed using the output fields from a level-coordinate GCM. In a level-coordinate GCM, there is no prognostic equation for MLT, so the MLT budget has to be evaluated diagnostically. In comparison, bulk mixed layer models (e.g., Chang 1994; Zhang et al. 2004) or isopycnic models (Wallcraft et al. 2003) employ a prognostic equation for MLT, thus closing the MLT budget by construction. A similar attempt to close the MLT budget using the output of a level-coordinate GCM was made by Vialard and Delecluse (1998), but they suggested that the entrainment heat advection cannot be written in discrete equations. Here we will demonstrate that the discrete equations may be derived.

Our entrainment schemes are tested between 1997 and 2000 in three areas (Fig. 1): the north-central Pa-

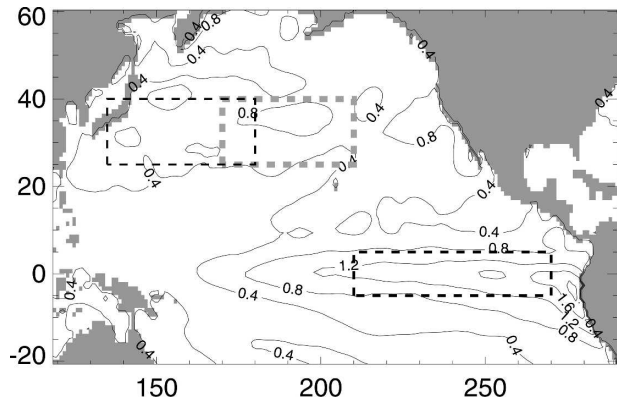


FIG. 1. The study areas as indicated by the rectangles: the north-central Pacific (25° – 40° N, 170° – 210° E), the Kuroshio Extension (25° – 40° N, 135° E– 180°), and the Niño-3 region (5° S– 5° N, 150° – 90° W) areas. The background fields are the standard deviation of interannual variability in Reynolds SST in K.

cific (25° – 40° N, 170° – 210° E), the Kuroshio Extension (25° – 40° N, 135° E– 180°), and the Niño-3 region (5° S– 5° N, 150° – 90° W). The time frame covers the 1997/98 El Niño and the abrupt warming around 1998 in the north-central Pacific (Kim et al. 2004). Parts of the interannual warming in the Kuroshio Extension area are also captured, which occurred from mid-1996 until mid-1998 (Vivier et al. 2002). The north-central Pacific is defined as the area with the largest SST change between 1997 and 2000 in the North Pacific.

2. The GCM solution

Model results used for this investigation are obtained from a data assimilation product of the Consortium for Estimating the Circulation and Climate of the Ocean (ECCO; more information is available online at <http://www.ecco-group.org>; see also Stammer et al. 2002). The model used is the parallel version of the primitive equation Massachusetts Institute of Technology (MIT) Ocean GCM (Marshall et al. 1997). The spatial domain is nearly global (80° S– 80° N). The model has a uniform resolution of $1^{\circ} \times 1^{\circ}$ except within 20° of the equator, in which meridional grid spacing is gradually reduced to 0.3° within 10° of the equator. There are 46 vertical levels with layer thickness of 10 m in the upper 150 m and 21 layers above 300 m. The model employs two advanced mixing schemes: the K-profile parameterization (KPP) vertical mixing (Large et al. 1994) and the Gent–McWilliams (GM) isopycnal mixing (Gent and McWilliams 1990). The model is forced by the National Centers for Environmental Prediction (NCEP) reanalysis products (12-hourly wind stress, daily diabatic air–

sea fluxes) with the time-means replaced by those of the Comprehensive Ocean–Atmosphere Data Set fluxes (da Silva et al. 1994). In addition to this imposed heat flux, model SST is relaxed to NCEP’s SST analysis. The model was first spun up for 10 yr from the rest using climatological temperature and salinity (Boyer and Levitus 1998) forced by seasonal climatological forcings averaged from 1980 to 1997. An approximate Kalman filter and smoother (Fukumori 2002) are used to assimilate anomalies of sea level and subsurface temperature obtained from the TOPEX/Poseidon altimeter and the Global Telecommunication System (D. Behringer 2002, personal communication), respectively. The mixed layer is defined diagnostically from the GCM output fields, such that at the base of the mixed layer the density is larger than that at 5 m by 0.125 kg m^{-3} . The density offset corresponds to a 0.5-K temperature offset at 35 psu and 20°C , which is used commonly for temperature profiles in the Tropics (e.g., Kessler et al. 1998). The budget closure schemes developed herein may be applied to level-coordinate GCM fields with other definitions of mixed layer depth.

3. Definition of the MLT budget closure and identification of a problem

Let us start by establishing the relationship between entrainment heat advection and the MLT balance. The balance of temperature (T) can be expressed as

$$\frac{\partial T_{\text{prognostic}}}{\partial t} = \frac{1}{\rho C_p} \frac{\partial q}{\partial z} - \nabla \cdot (\mathbf{v}T) + \text{mixing}, \quad (1)$$

where q is penetrative shortwave flux, ρC_p the density times specific heat of seawater, $\mathbf{v} = (u, v, w)$ the three-dimensional velocity, $\nabla = (\partial/\partial x, \partial/\partial y, \partial/\partial z)$, and “mixing” denotes the turbulent mixing. The “mixing” in our model refers to the sum of $\nabla_z(\kappa_z \nabla_z T)$, the KPP nonlocal component, and the GM mixing, where $\nabla_z = \partial/\partial z$ and κ_z is the turbulent diffusivity in vertical direction. Equation (1) gives a prognostic formula for the temperature tendency ($\partial T_{\text{prognostic}}/\partial t$). The depth average of the rhs of Eq. (1) from the surface to the mixed layer depth (h) at a single horizontal grid point yields the diagnostic balance of MLT (similar to Vialard and Delecluse (1998); also the derivation without advection is available in appendix A)

$$\begin{aligned} \frac{\partial [T_{\text{diagnostic}}]}{\partial t} &= \left[\frac{1}{\rho C_p} \frac{\partial q}{\partial z} \right] + \frac{1}{h} (\kappa_z \nabla_z T)_{z=0} - [\nabla_H \cdot (uT, vT)]_{\text{ml}} + [\text{ML mixing}] + \text{subsurface} \\ \text{subsurface} &= -\frac{1}{h} \Delta T \frac{\partial h}{\partial t} - [\nabla_H \cdot (uT, vT)]_{\text{induct}} - [\nabla_z (wT)] - \frac{1}{h} (\kappa_z \nabla_z T)_{z=-h}, \end{aligned} \quad (2)$$

where the square bracket represents the depth average within the mixed layer and $\nabla_H = (\partial/\partial x, \partial/\partial y)$. Here $(\kappa_z \nabla_z T)_{z=0}$ is the sum of outgoing longwave, sensible, and latent heat fluxes. The horizontal advective tendency is divided into one occurring within the mixed layer and the “lateral induction” resulting from the horizontal advection across a sloping mixed layer base, $[\nabla_H \cdot (uT, vT)]_{\text{ml}}$ and $[\nabla_H \cdot (uT, vT)]_{\text{induct}}$ respectively. The “ML mixing” is the tendency due to the mixing within the mixed layer, and represents the GM mixing and the KPP nonlocal component. The subsurface processes [denoted “subsurface” in Eq. (2)] consist of the entrainment heat advection and the vertical mixing at the mixed layer base. The entrainment heat advection is the sum of the temporal mixed layer depth variation ($\Delta T \partial h / \partial t$), the lateral induction, and the vertical advection (e.g., Cushman-Roisin 1987). Here $[\nabla_z (wT)]$ contains a component at the air–sea boundary, which is not a subsurface process. However, its size is at most two orders of magnitude smaller than its subsurface counterpart.

As a first step in analyzing the MLT budget, we

verify our calculation by examining budget closure of a layer of fixed depth, in which $\partial h / \partial t$ is zero. All the terms in the temperature budget in Eq. (1) are computed at the model’s integration time step and archived as 30-day averages. The temperature budget closes exactly (not shown).

For the mixed layer with variable depths, we find that the budget does not close if $\Delta T \partial h / \partial t$ is omitted (not shown) or if it is evaluated using ad hoc schemes. The lateral induction and the vertical advection are advection processes and thus the entrainment heat advection by these two elements is determined unambiguously by the GCM. Here we consider three ad hoc schemes used in previous studies to evaluate the temperature of the entrained water. One is to use temperatures from a layer below the mixed layer ($T_{-h, \text{layer}}$; similar to Alexander et al. 2000). The second one is to sample the entrainment temperature at a particular depth below the mixed layer base [20 m below the mixed layer base by Yasuda et al. (2000) for the Kuroshio Extension and by McPhaden (2002) for the equatorial Pacific, and 10 m below by Qu (2003) for the subtropical North

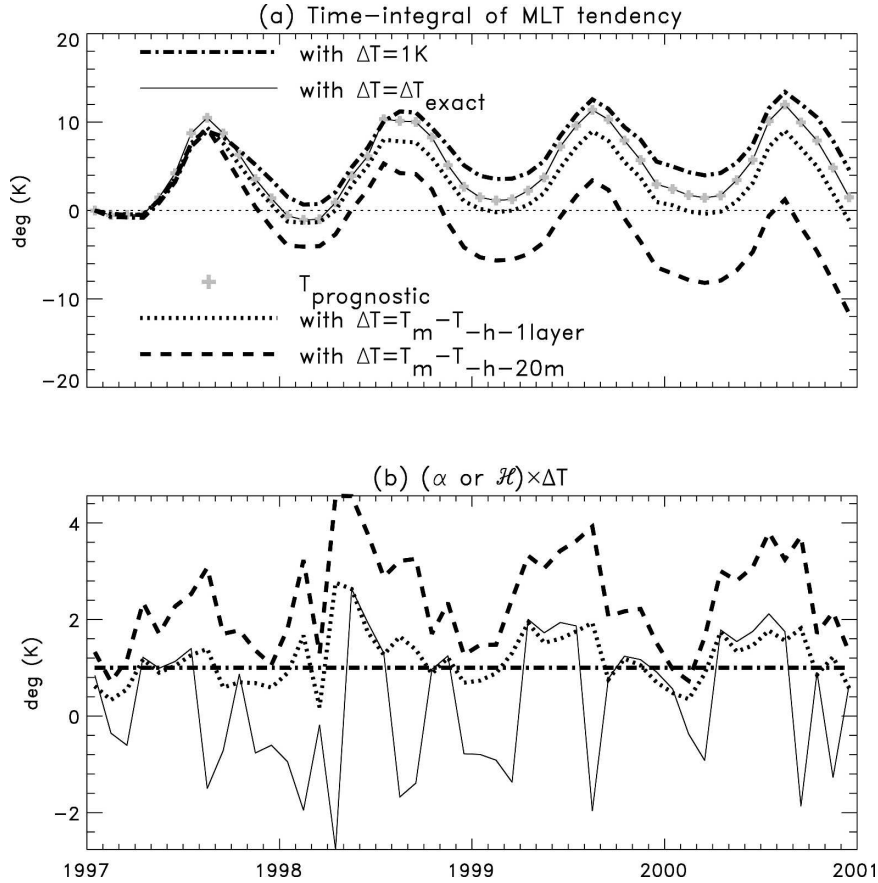


FIG. 2. (a) Examining the closure of the MLT tendency budget at 35.5°N , 169.5°W by comparing the prognostic MLT with the diagnostic MLT evaluated using four formulations of the temperature difference (ΔT) between mixed layer water and entrained water. (b) The corresponding $\alpha\Delta T_{\text{exact}}$, $\mathcal{H}\Delta T_{1\text{layer}}$, and $\mathcal{H}\Delta T_{20\text{m}}$ where α is -1 for entrainment and $+1$ for detrainment, and $\mathcal{H}$ is -1 for entrainment and 0 for detrainment. The time mean of $\alpha\Delta T_{\text{exact}}$, $\mathcal{H}\Delta T_{1\text{layer}}$, and $\mathcal{H}\Delta T_{20\text{m}}$ are 0.93 , 1.59 , and 2.79 K, respectively.

Pacific]. Here we simulate this second approach by sampling the model at 20 m below the mixed layer base ($T_{h-20\text{m}}$). The third is to use a constant temperature for ΔT [1 K by Qiu and Kelly (1993) for the Kuroshio Extension and 0.75 K by Swenson and Hansen (1999) for the equatorial Pacific]. Figure 2 compares the MLT reconstructed from these tendency estimates with the actual model's MLT ($T_{\text{prognostic}}$) at an arbitrarily chosen location in the north-central Pacific (35.5°N , 169.5°W). Not surprisingly, none of the ad hoc schemes for ΔT closes the budget. An exact formulation to evaluate $\Delta T \partial h / \partial t$ is derived in the following section.

4. Exact formulation of the MLT budget at a single location

Let us consider a time series of a temperature–depth profile from a GCM at a particular horizontal location

where the mixed layer depth varies in time. Let the temperature at a depth level k at time t be $T_k(t)$. From Eq. (1) and Fig. 3,

$$T_k(t) = T_k(t-1) + \text{ST}_k(t), \quad (3)$$

where ST is the sum of tendency integrated over time. For an entrainment case (Fig. 3a) the mixed layer depth increases from h_1 at $t-1$ to h_1+h_2 at t , where h_k is the thickness of layer k . Then we obtain

$$T_{\text{ml}}(t) = (h_1 T_1(t-1) + h_2 T_2(t-1)) / H + \text{ST}_{\text{ml}}(t), \quad (4)$$

$$T_{\text{ml}}(t-1) \equiv T_1(t-1), \quad (5)$$

where $H = h_1 + h_2$ and ml denotes the mixed layer. Here T_1 and T_2 are average temperatures over layers h_1 and h_2 , respectively. Here $\text{ST}_{\text{ml}}(t)$ is the tendency of the mixed layer defined at time t and is

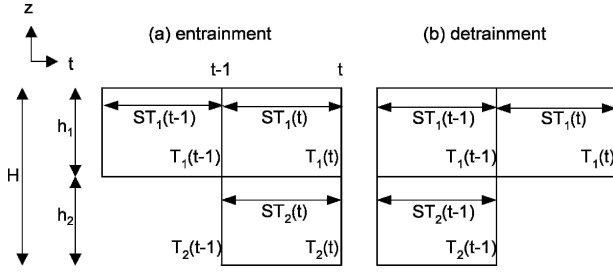


FIG. 3. Discrete formulation of entrainment tendency by the temporal mixed layer variation term at one horizontal grid point for the (a) entrainment and (b) detrainment cases. Here ST stands for the sum of the temperature tendency components on the rhs of Eq. (1) after integrating over time. In $T_k(t)$, T is the temperature, and the subscript k and the argument t denote the depth and time indexes, respectively. Horizontal and vertical axes represent time (t) and depth (z). The top and bottom boundaries of the water column correspond to the sea surface and the mixed layer base, respectively. Note that $T(t)$ is an instantaneous temperature at time t .

$(h_1ST_1(t) + h_2ST_2(t)) / H$. Then, substituting Eq. (5) into Eq. (4) yields

$$T_{ml}(t) = T_{ml}(t-1) + ST_{ml}(t) - \frac{h_2}{H} (T_1(t-1) - T_2(t-1)). \quad (6)$$

A similar analysis yields a formula for a detrainment case (Fig. 3b) as

$$T_{ml}(t) = T_{ml}(t-1) + ST_{ml}(t) + \frac{h_2}{H} (T_1(t-1) - T_2(t-1)), \quad (7)$$

where now $ST_{ml}(t) = ST_1(t)$. The general discrete equation for the MLT budget may be written as

$$T_{ml}(t) = T_{ml}(t-1) + ST_{ml}(t) + \alpha \frac{\Delta h}{H} (T_1(t-1) - T_2(t-1)), \quad (8)$$

where T_1 denotes the depth-mean temperature of the layer(s) that stays within the mixed layer between $t-1$

and t . Here T_2 denotes the depth-mean temperature of the layer(s) that is about to be entrained to or detrained from the mixed layer from $t-1$ to t . Here Δh is the thickness of this layer(s) that is about to be entrained/detrained and is positive for both entrainment and detrainment cases, and H is the larger of the mixed layer depth from $t-1$ to t . Here α is a coefficient defined as

$$\alpha \equiv \begin{cases} -1 & \text{when water is entrained} \\ 0 & \text{when no change in depth} \\ 1 & \text{when water is detrained} \end{cases}.$$

In comparison to the ad hoc schemes discussed at the end of section 3, the MLT budget closes exactly using the last term of Eq. (8) for $\Delta T \partial h / \partial t$ (Fig. 2). Hereafter the MLT budget equation that can close the budget exactly will be called the *exact* formulation.

We now assess the differences between the exact formulation and the three ad hoc formulations used in the past. The MLT balance given by the three ad hoc formulations may be expressed in a general form:

$$T_{ml}(t) = T_{ml}(t-1) + ST_{ml}(t) + \mathcal{H} \frac{\Delta h}{H'(t)} (T'_{ml}(t-1) - T_{ent}(t-1)). \quad (9)$$

Here $\mathcal{H}$ is the Heaviside function, defined as -1 during entrainment and 0 otherwise. The differences between the exact and the past formulations lie in $\mathcal{H}$, H' , T'_{ml} , and T_{ent} , and are summarized in Table 1. The prime symbol distinguishes the variables used in the ad hoc formulations from those of the exact formulation. Figure 4 shows the differences in terms of H , ΔT , and $\Delta T \partial h / \partial t$. For the exact formulation, $\Delta T_{\text{exact}} \equiv T_1 - T_2$ and for the ad hoc formulations, $\Delta T_{1K} \equiv 1 \text{ K}$, $\Delta T_{1\text{layer}} \equiv T'_{ml} - T_{-h-1\text{layer}}$, and $\Delta T_{20m} \equiv T'_{ml} - T_{-h-20m}$. Important results from Fig. 4 are as follows.

- The exact formulation of the entrainment heat advection occasionally contributes to warming the MLT (the black curve in the bottom row being sometimes positive). This warming occurs during the mixed layer

TABLE 1. Differences between the exact and the past ad hoc formulations of the entrainment tendency by temporal mixed layer depth variation. The tabulated parameters are defined in Eqs. (8) and (9).

Parameter	Exact formulation	Past formulations
α or $\mathcal{H}$ during detrainment	+1	Zero
Δh	The same	
H or H'	$\text{Max}(h(t-1), h(t))$	$h(t)$
T_1 or T'_{ml}	Depth average of $T(t-1)$ from the sea surface to $\min(h(t-1), h(t))$	Depth average of $T(t-1)$ from the sea surface to $h(t-1)$
T_2 or T_{ent}	Depth average of $T(t-1)$ between $h(t-1)$ and $h(t)$	Various ad hoc formulations

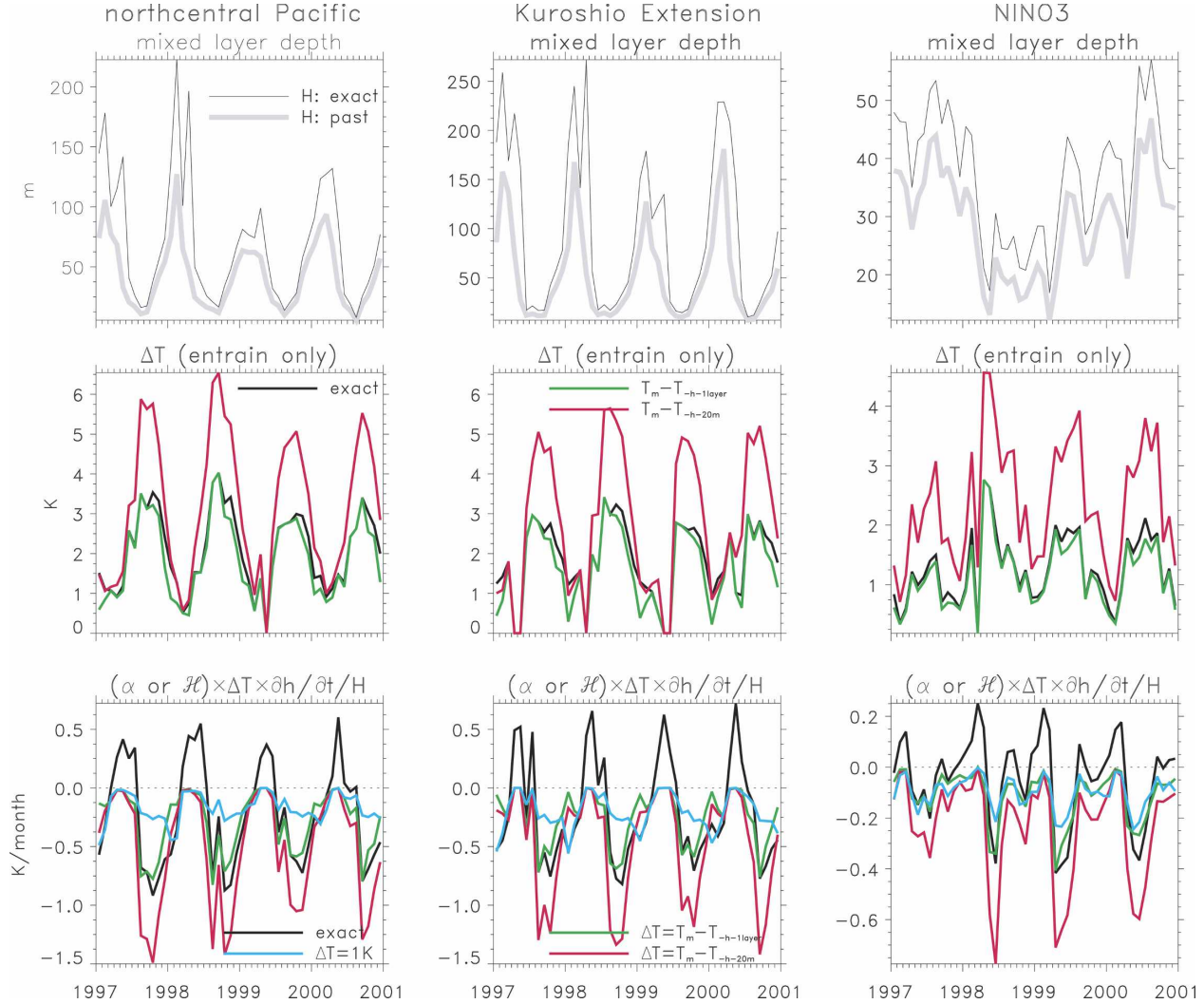


FIG. 4. Entrainment tendency by the temporal mixed layer variation term ($\Delta T \partial h / \partial t / H$) at a single grid point: (top row) mixed layer depth (H and H'), (middle row) mixed layer temperature minus entrainment temperature (ΔT), and (bottom row) entrainment tendency [$\alpha \Delta T \partial h / \partial t / H$ and $\mathcal{H} \Delta T \partial h / \partial t / H'$, defined in Eqs. (8) and (9)]. Columns correspond to three study regions. Each record in the time series plotted is an area-weighted average of the values at a single grid point. Here ΔT is zero when there is no entrainment (e.g., May 1999 in the north-central Pacific). Exact refers to the entrainment heat advection that enables the MLT budget closure.

shoaling because of the small vertical temperature gradient within the mixed layer. The mixed layer shoaling can cause slightly colder water to exit the mixed layer, which effectively increases the average temperature of the mixed layer after the shoaling. This finding is not inconsistent with the zero heat advection during the shoaling by imposing the Heaviside function when the mixed layer is treated as a bulk layer, because then the vertical temperature gradient within the mixed layer is zero (e.g., Kraus and Turner 1967; Niiler and Kraus 1977). To close the budget, it is necessary to take into account the warming tendency during the mixed layer shoaling.

- The H of the exact formulation is identical to $H'(t)$ of the past formulations during the mixed layer deepening and is greater than $H'(t)$ during shoaling (Table 1). Therefore, the largest difference between H and H' is found in the shoaling season (late winter and spring) in the north-central Pacific and the Kuroshio Extension (Fig. 4). The difference between H and H' in Fig. 4 is not exactly zero during the deepening season because shoaling occurs only at limited locations and the figure shows an areal average. In the Niño-3 area the seasonal cycle is not as strong as in midlatitudes, and the mixed layer shoaling may occur throughout a year. Consequently, the difference be-

TABLE 2. Comparison of ΔT between the exact and the ad hoc formulations at a single grid point. The minimum, time-mean, and maximum values are compiled from the time series plotted in the middle row in Fig. 4. The comparison is made for entrainment cases only. Units are in K.

	Min	Mean	Max
North-central Pacific			
$\Delta T_{\text{exact}} - \Delta T_{1\text{layer}} (= T_{-h,1\text{layer}} - T_2)$	0	0.22	0.92
$\Delta T_{\text{exact}} - \Delta T_{20\text{m}} (= T_{-h,20\text{m}} - T_2)$	-2.51	-1.07	0.04
Kuroshio Extension			
$\Delta T_{\text{exact}} - \Delta T_{1\text{layer}}$	0	0.24	0.93
$\Delta T_{\text{exact}} - \Delta T_{20\text{m}}$	-2.66	-0.95	0.34
Niño-3			
$\Delta T_{\text{exact}} - \Delta T_{1\text{layer}}$	0	0.10	0.35
$\Delta T_{\text{exact}} - \Delta T_{20\text{m}}$	-1.96	-1.10	-0.36

tween H and H' in the Niño-3 area does not exhibit strong seasonality.

- The ΔT of the exact formulation, ΔT_{exact} , is smaller than those of the ad hoc formulations in the past by as much as a factor of 2 (Fig. 4). The comparison is limited to entrainment cases only, because the past studies set ΔT to zero during detrainment. The MLTs of the exact and the past formulations (T_1 and T_{ml}) are identical during entrainment, and therefore the differences in ΔT originate entirely from the temperatures of entrained water (T_2 , $T_{-h,1\text{layer}}$, and $T_{-h,20\text{m}}$). $T_{-h,1\text{layer}}$ is slightly higher than T_2 (Fig. 4 and Table 2), because the entrained layers can be more than one layer. Here $T_{-h,20\text{m}}$ is substantially lower than T_2 in midlatitude summer (Fig. 4), because 20 m below a mixed layer base for $T_{-h,20\text{m}}$ corresponds to two vertical layers in the GCM used but Δh for T_2 is usually one layer in this season. Strong stratification in midlatitude summer is another cause of the large value of $T_{-h,20\text{m}} - T_2$ (for the same reason, $T_{-h,20\text{m}} - T_2$ in the Niño-3 area is large throughout a year). In midlatitude winter, Δh is generally larger than 20 m and $T_{-h,20\text{m}} - T_2$ can be positive (see the maximum of the difference in Table 2).
- The consequent contribution by $\Delta T \partial h / \partial t$ to the MLT budget varies significantly depending on which formulation is used (Fig. 4). As a result of the large $\Delta T_{20\text{m}}$, this definition for ΔT gives the largest entrainment tendency. By comparison, the entrainment tendencies given by $\Delta T_{1\text{layer}}$ and by $\Delta T_{1\text{K}}$ agree better with that by ΔT_{exact} . However, large discrepancy is found during the detrainment periods.

5. Exact formulation of the MLT budget over a region

To understand the MLT balance over a large area, a volume-weighted mean of MLT is a more appropriate

quantity than a commonly used area-weighted mean. First of all, the volume mean of MLT represents more effectively the ocean mixed layer's capacity to exchange heat with the atmosphere than the area mean does. With the area weights, for example, regions with deep mixed layers will contribute to the mean MLT with the same weights as those of the shallow mixed layer regions. This will misleadingly reduce the relative capacity of the deep mixed layers in air-sea heat exchange. It should be borne in mind, however, that the area- and volume-mean MLT are closely related quantities. They correlate with each other highly (>0.98 in the three areas of interest of this study), while they show discrepancy in the magnitude of the variability (by about 20%). The second reason for employing the volume-weighted MLT is that the area mean of the local MLT balance at different horizontal grid points may distort the contributions by advective tendencies. For example, when the local temperature balance is averaged over the Niño-3 region using area weights, the zonal advection has a cooling effect during the El Niño period, contrary to conventional expectation (Lee et al. 2004). Lee et al. demonstrated that the correct characterization of the zonal advective tendency may be obtained by evaluating the advective contribution to the volume-mean temperature. An example of a complete MLT budget analysis using the volume-weights MLT is available in Kim et al. (2004), where we studied the 1997–2000 interannual warming in the upper ocean of the north-central Pacific.

The volume-weighted average of the MLT balance at

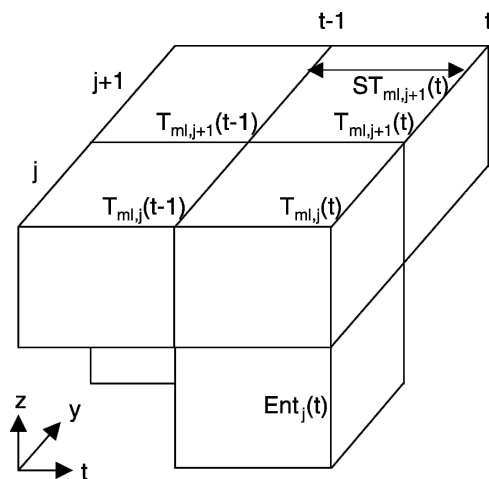


FIG. 5. Discrete formulation of Ent_volume for a meridional strip of a mixed layer. At j the mixed layer is deepening and at $j + 1$, it is shoaling. The subscript ml denotes the depth average over the mixed layer at one horizontal grid point. Other symbols are the same as in Fig. 3.

one horizontal grid point, Eq. (8), does not describe the changes in the volume-weighted mean MLT. That is, $\int_{V(t)} (T(t) - T(t-1))dV$ from Eq. (8) is different from the change of the volume-mean MLT, $\int_{V(t)} T(t)dV - \int_{V(t-1)} T(t-1)dV$. This is because of the change of mixed layer volume, $V(t)$, which is the volume of the mixed layer over the entire region at time t . The discrepancy arises because a volume-mean value should be used for T_1 during the evaluation of ΔT , instead of only a depth-averaged value as in Eq. (6). An analytical equation of the volume-mean MLT budget is presented in appendix A. However, the discrete form of the analytical equation is complicated. Here we demonstrate a simpler form of the discrete MLT balance equation. Let us consider a simple case of a meridional strip of a mixed layer in the y - z plane that consists of two y grid cells (Fig. 5). Let the mixed layer deepen in time at $y = j$ and shoal at $y = j + 1$ to simulate the general case of both entrainment and detrainment. The following derivation may be expanded to a mixed layer in the x - y - z volume of any size without loss of generality. From the exact form of the MLT budget at one location in Eq. (8),

$$T_{\text{ml},j}(t) = T_{\text{ml},j}(t-1) + \text{ST}_{\text{ml},j}(t) + \text{Ent}_j(t),$$

$$T_{\text{ml},j+1}(t) = T_{\text{ml},j+1}(t-1) + \text{ST}_{\text{ml},j+1}(t) + \text{Ent}_{j+1}(t), \quad (10)$$

where $T_{\text{ml},j}(t)$ is the depth-averaged mixed layer temperature at $y = j$ and at time t , and the entrainment heat advection Ent is

$$\text{Ent}_j(t) \equiv \alpha \frac{\Delta h_j}{H_j} (T_{1,j}(t-1) - T_{2,j}(t-1)).$$

The mean MLT over the entire volume is

$$\langle T_{\text{ml}}(t) \rangle = \frac{w_{j+1}(t)T_{\text{ml},j+1}(t) + w_j(t)T_{\text{ml},j}(t)}{w_{j+1}(t) + w_j(t)} \quad (11)$$

where the angle brackets denote a volume-weighted average over the mixed layer volume and $w_j(t)$ denotes the volume of the mixed layer at location j and time t . Substitution of Eq. (10) into Eq. (11) gives

$$\langle T_{\text{ml}}(t) \rangle = \langle \text{ST}_{\text{ml}}(t) \rangle + \langle \text{Ent}(t) \rangle + \frac{w_{j+1}(t)T_{\text{ml},j+1}(t-1) + w_j(t)T_{\text{ml},j}(t-1)}{w_{j+1}(t) + w_j(t)},$$

which can be rewritten as

$$\langle T_{\text{ml}}(t) \rangle = \langle T_{\text{ml}}(t-1) \rangle + \langle \text{ST}_{\text{ml}}(t) \rangle + \langle \text{Ent}(t) \rangle + \text{Ent_volume}(t), \quad (12)$$

where “Ent_volume” represents the correction to entrainment heat advection to describe the effect of the changes in volume-mean MLT and is given as

$$\begin{aligned} \text{Ent_volume}(t) \equiv & \frac{w_{j+1}(t)T_{\text{ml},j+1}(t-1) + w_j(t)T_{\text{ml},j}(t-1)}{w_{j+1}(t) + w_j(t)} \\ & - \frac{w_{j+1}(t-1)T_{\text{ml},j+1}(t-1) + w_j(t-1)T_{\text{ml},j}(t-1)}{w_{j+1}(t-1) + w_j(t-1)}. \end{aligned} \quad (13)$$

Ent_volume is nonzero only if mixed layer depth varies in both space and time. To understand Ent_volume, let us take an example of the north-central Pacific. Let $w_{j+1}(t)$ and $w_j(t)$ represent the mixed layer depth in the north and south within the domain. From summer to winter, $w_{j+1}(t)$ increases faster than $w_j(t)$ as a result of stronger convective cooling in the north. The differential rate of the mixed layer depth change with respect to the latitude is presented in Fig. 6. Consequently, the center of volume shifts to the north. Such northward movement of the center means the increase in the proportion of the northern water within the mixed layer.

As a result the volume-mean MLT over the entire mixed layer will decrease because the northern water is generally colder than the southern water. These processes will make the value of the first term in rhs of Eq. (13) smaller than that of the second term, producing a cooling effect. From winter to summer the center of volume moves southward, leading to a warming effect. The cooling and warming trends are clearly visible in Fig. 7 that shows the time series of Ent_volume. The same figure further shows that in midlatitudes the size of Ent_volume is large. Ent_volume in midlatitudes cools the mixed layer by about 3 K until the end of

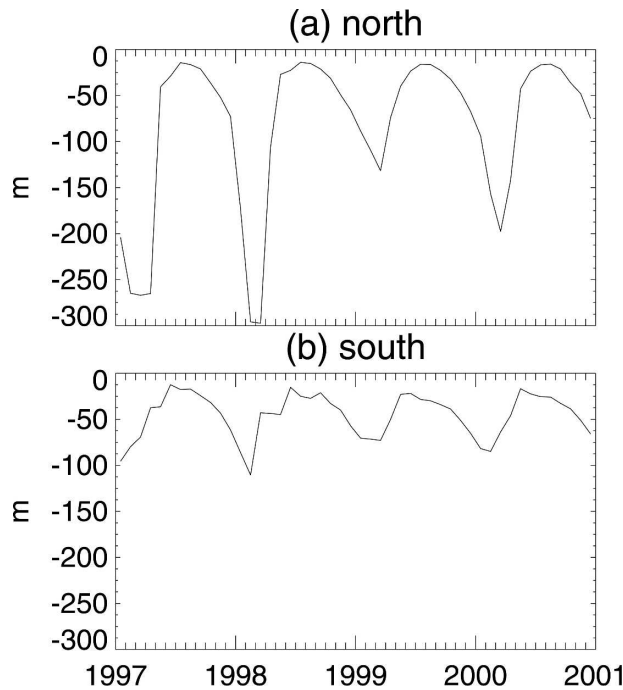


FIG. 6. Spatial variation of mixed layer depth in the north-central Pacific area. The depths are averaged over (a) 37°–40°N, 170°–210°E and (b) 25°–28°N, 170°–210°E.

winter (~February) and warms by a larger amount from the beginning of spring. In the Niño-3 area, however, *Ent_volume* is a minor term because the size of the temporal variation in mixed layer depth is fairly uniform in space. For example, $\delta\text{MLD}(t)|_{\text{lat } 5^{\circ}\text{N}} \approx \delta\text{MLD}(t)|_{\text{lat } 5^{\circ}\text{S}}$, where δMLD denotes the annual variation of the mixed layer depth. In the north-central Pacific, $\delta\text{MLD}(t)|_{\text{lat } 40^{\circ}\text{N}} \gg \delta\text{MLD}(t)|_{\text{lat } 25^{\circ}\text{N}}$, leading to the large *Ent_volume*.

Equation (12) is the exact form of the MLT budget over a region. The consequent precise closure of the MLT budget is presented in Fig. 8. Appendix B provides practical information on the fields that a GCM needs to produce to evaluate the exact forms of the MLT balance. Below we assess the impact of the exact formulations on the volume-mean MLT balance.

The imbalance in the annual MLT budget associated with the ad hoc formulations by ΔT_{1K} , $\Delta T_{1\text{layer}}$, and ΔT_{20m} is about 0.7, 0.7, and 3 K yr⁻¹, respectively (Table 3). For the north-central Pacific and the Kuroshio Extension, the imbalance due to the use of ΔT_{1K} and $\Delta T_{1\text{layer}}$ schemes may be considered small but that due to the use of ΔT_{20m} is large, considering that the annual peak-to-peak variation of MLT is about 10 K yr⁻¹ (Figs. 8a,b). For the Niño-3 volume, the imbalance is relatively more significant considering the small magnitude of the annual variation (about 5 K yr⁻¹, Fig. 8c).

The imbalance is equivalent to an annual mean heat flux of 13–35 W m⁻² in midlatitudes and 6–17 W m⁻² in the Niño-3 region (Table 3).

Ent_volume warms the mixed layer in all the three regions (Fig. 7), compensating for approximately 50% of the cooling by $\Delta T \partial h / \partial t$. Thus without *Ent_volume*, $\Delta T \partial h / \partial t$ would result in too much cooling by the size of *Ent_volume* that is about 1 K yr⁻¹ in the north-central Pacific and Kuroshio Extension, and by about 0.3 K yr⁻¹ in the Niño-3 area.

6. Summary

We have developed a rigorous formulation of the entrainment heat advection to study the mixed layer temperature (MLT) balance using a level-coordinate GCM output. Because the developed formulation closes the MLT budget exactly, we refer to it as the *exact* formulation. The budget closure is verified using the output of the GCM for the period of 1997–2000 over the north-central Pacific, the Kuroshio Extension, and the Niño-3 area in the tropical Pacific.

The integral elements of the exact formulation are as follows: 1) the discrete formulation of the temperature difference (ΔT) between mixed layer water and entrained water to evaluate the temporal mixed layer variation component of entrainment heat advection ($\Delta T \partial h / \partial t$) at each horizontal grid point, 2) the additional entrainment heat advection term that is needed to close the budget for the volume-mean MLT balance (*Ent_volume*), and 3) the equivalent warming of the mixed layer during the mixed layer shoaling. The *Ent_volume* represents that ΔT is the difference between volume-mean MLT and entrainment temperature. The *Ent_volume* is nonzero only when the study area covers more than one horizontal grid point and only if mixed layer depth varies in both space and time. The warming tendency during the mixed layer shoaling results from shedding slightly colder water in the lower mixed layer, which effectively increases the averaged temperature of the mixed layer after the shoaling. Such warming has been assumed to be zero in bulk mixed layer models in the past, but is nonzero in GCMs and in reality because of the presence of small vertical temperature gradient in the mixed layer. The lateral induction and the vertical advective components of the entrainment heat advection are determined unambiguously by the GCM as the model's advective processes.

When comparing the exact formulation of $\Delta T \partial h / \partial t$ with its ad hoc estimation schemes used in the past studies, the imbalance in the MLT budget by the ad hoc schemes reaches a maximum of about 3 K yr⁻¹ or an annual mean of 6–35 W m⁻² in the three areas that are

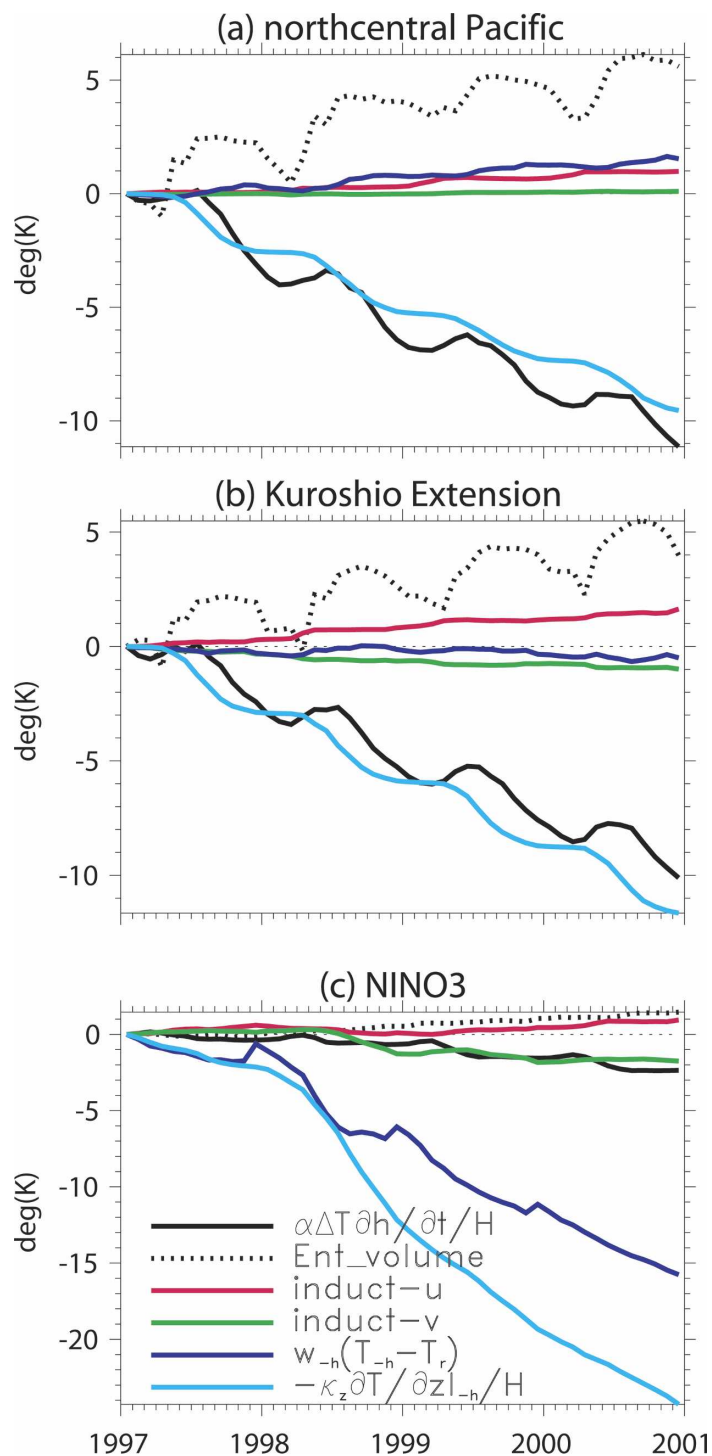


FIG. 7. Components of the subsurface tendencies contributing to the volume-mean MLT [defined in Eqs. (2) and (13)]. The tendency components are integrated in time: (a) north-central Pacific, (b) Kuroshio Extension, and (c) Niño-3. Induct- u and - v are the lateral induction caused by the horizontal advection across a sloping mixed layer base ($-\nabla_H \cdot (uT, vT)$). Here $w_{-h}(T_{-h} - T_r)$ is the tendency by the vertical advection through the mixed layer base at $z = -h$ and T_r denotes the reference temperature averaged over the mixed layer volume.

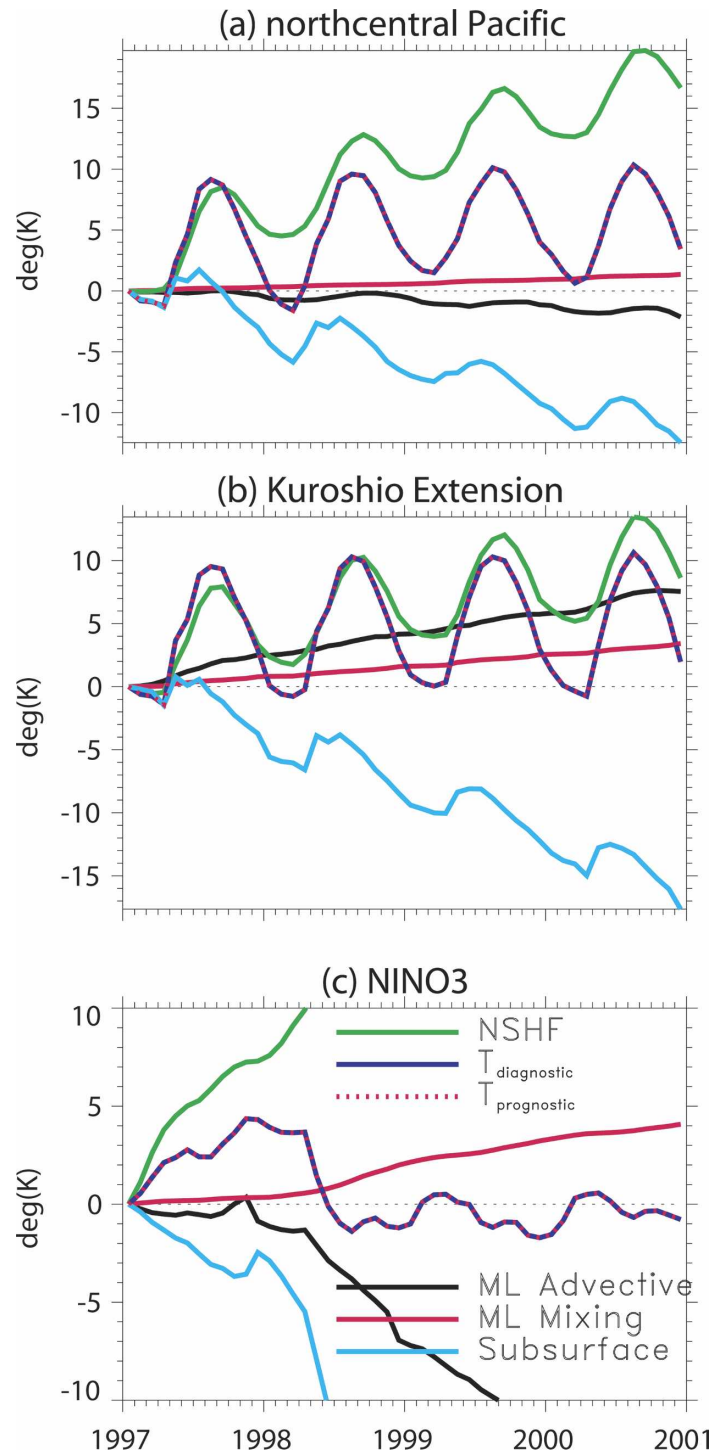


FIG. 8. Time-integrated MLT tendency over a volume and its components: (a) north-central Pacific, (b) Kuroshio Extension, and (c) Niño-3. Here $T_{\text{diagnostic}}$ (solid blue) is derived from the diagnostic MLT balance and the $T_{\text{prognostic}}$ (red dash) given by the prognostic GCM simulation are identical to each other. Net surface heat flux (NSHF) is the sum of longwave and turbulent fluxes at the sea surface and penetrative shortwave flux. The ML advective and ML mixing denote the tendencies due to advection and mixing within the mixed layer. In (c) the ordinate scale is selected to highlight the MLT change.

TABLE 3. Four-year mean of the imbalance in the volume-mean MLT budget, caused by the ad hoc formulations, is evaluated as the difference in the MLT between the exact formulation and the ad hoc formulation. The differences are shown in terms of K yr^{-1} and of equivalent W m^{-2} (in parentheses). The MLT tendency values are consistent with the time integral of the values presented in the bottom row in Fig. 4.

$\alpha \Delta T_{\text{exact}} \partial h / \partial t / H$ ($\rho C_p \alpha \Delta T_{\text{exact}} \partial h / \partial t$) minus	North-central Pacific	Kuroshio Extension	Niño-3
$\mathcal{H} \Delta T_{1\text{K}} \partial h / \partial t / H'$ ($\rho C_p \mathcal{H} \Delta T_{1\text{K}} \partial h / \partial t$)	-0.7 (21.0)	-0.1 (34.4)	0.7 (5.8)
$\mathcal{H} \Delta T_{1\text{layer}} \partial h / \partial t / H'$ ($\rho C_p \mathcal{H} \Delta T_{1\text{layer}} \partial h / \partial t$)	0.7 (13.0)	0.5 (16.9)	0.8 (6.0)
$\mathcal{H} \Delta T_{20\text{m}} \partial h / \partial t / H'$ ($\rho C_p \mathcal{H} \Delta T_{20\text{m}} \partial h / \partial t$)	3.6 (34.7)	3.1 (35.3)	2.3 (17.4)

considered. The Ent_volume warms the mixed layer and compensates for about half of the cooling by $\Delta T \partial h / \partial t$. The negligence of Ent_volume would further introduce an imbalance of a maximum of 1 K yr^{-1} . Such imbalance due to $\Delta T \partial h / \partial t$ and Ent_volume is significant, since the annual peak-to-peak MLT variation is about 10 K yr^{-1} in the north-central Pacific and the Kuroshio Extension, and the variation is even smaller in the Niño-3 area.

Finally, the developments from this study may apply to the MLT budget analysis using observational data. Although exact closure of the budget would be difficult due to the incomplete sampling by observations, the incorporation of the exact form of ΔT , the warming tendency by the mixed layer shoaling, and Ent_volume would improve the accuracy of the estimated entrainment effect. Also the developments may be used for a model with different grid sizes such as one with higher depth resolution within a mixed layer.

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APPENDIX A

Analytical Equation for the Volume-Mean MLT Budget

Here we derive an analytical equation describing the change of the volume-mean MLT. Let us start with the MLT balance at one horizontal grid point. We set advective tendencies to zero for simplicity, because they may be added linearly to the balance equations:

$$\frac{\partial T}{\partial t} = \frac{1}{\rho C_p} \frac{\partial q}{\partial z} + \text{mixing}, \quad (\text{A1})$$

where the symbols are the same as in Eq. (1). Vertical integration over the column of the mixed layer yields

$$\frac{\partial}{\partial t} h[T] - T_{-h} \frac{\partial h}{\partial t} = \int_{-h}^0 \left(\frac{1}{\rho C_p} \frac{\partial q}{\partial z} + \text{mixing} \right) dz, \quad (\text{A2})$$

where the symbols are the same as in Eq. (2). Equation (A2) may be rewritten into the equation for the change of MLT at a horizontal grid point:

$$\frac{\partial}{\partial t} [T] = \left[\frac{1}{\rho C_p} \frac{\partial q}{\partial z} + \text{mixing} \right] - \frac{1}{h} ([T] - T_{-h}) \frac{\partial h}{\partial t}. \quad (\text{A3})$$

Equation (A3) is identical to the discrete form, Eq. (8), with zero advective tendencies. Now let us consider the change of MLT over a volume such as the mixed layer of the Niño-3 area. Integrating Eq. (A2) over a volume, we obtain

$$\begin{aligned} \iint_S \left(\frac{\partial}{\partial t} h[T] - T_{-h} \frac{\partial h}{\partial t} \right) dx dy = \\ \int_V \left(\frac{1}{\rho C_p} \frac{\partial q}{\partial z} + \text{mixing} \right) dV, \end{aligned} \quad (\text{A4})$$

where V is the entire volume and S is its horizontal area. Here V is a function of time but S is time invariant. Therefore,

$$\begin{aligned} \int_S \int \frac{\partial}{\partial t} h[T] dx dy &= \frac{d}{dt} \int_V T dV = \frac{d}{dt} V \langle T \rangle \\ &= V \frac{d}{dt} \langle T \rangle + \langle T \rangle \frac{d}{dt} V \\ &= V \frac{d}{dt} \langle T \rangle + \iint_S \langle T \rangle \frac{\partial h}{\partial t} dx dy, \end{aligned} \quad (\text{A5})$$

where the angle brackets denote a volume-weighted average over the mixed layer volume. With Eq. (A5), Eq. (A4) becomes

$$V \frac{d}{dt} \langle T \rangle + \int_S \langle T \rangle - T_{-h} \frac{\partial h}{\partial t} dx dy = \int_V \left(\frac{1}{\rho C_p} \frac{\partial q}{\partial z} + \text{mixing} \right) dV,$$

which gives the analytical equation describing the change of volume-mean MLT:

$$\frac{d}{dt} \langle T \rangle = \left\langle \frac{1}{\rho C_p} \frac{\partial q}{\partial z} + \text{mixing} \right\rangle - \frac{1}{V} \int_S \langle T \rangle - T_{-h} \frac{\partial h}{\partial t} dx dy. \quad (\text{A6})$$

An important feature of Eq. (A6) is that it is the volume-mean MLT, $\langle T \rangle$, that characterizes the entrainment heat advection for the volume-mean MLT balance.

Comparing the volume-integrals of the point MLT budget equation in Eq. (A3) and the volume-mean budget equation in Eq. (A6), the discrepancy is found *only* in the mixed layer temperature determining the entrainment, that is, $\iint_S \langle T \rangle (\partial h / \partial t) dx dy \neq \int_V (1/h) [T] (\partial h / \partial t) dV$. From this we conclude that the incorrect choice of mixed layer temperature during entrainment calculation explains why simple volume-mean of the pointwise MLT balance does not describe the balance of the volume-mean MLT change.

To implement Eq. (A6) into a discrete form, we tried expanding the formulation of $\Delta T \partial h / \partial t$ at one horizontal grid point [Eq. (8), derived for the z - t domain shown in Fig. 3] into the y - z - t domain shown in Fig. 5. However, the expansion is rather complicated and is not easily simplified. In comparison, we have eliminated the z dimension by the depth-average that produces $T_{\text{ml},j}(t)$ in Eq. (10), which enabled the simplification into Eq. (13).

APPENDIX B

Requirements for Evaluating the MLT Budget

This section provides practical information on the fields that a GCM needs to produce to evaluate the exact forms of the MLT balance. Let the MLT budget be analyzed over a time interval of τ . First, the components of the prognostic MLT equation, Eq. (1), need to be archived at all the model grid locations. These components may be averaged over τ during the model integration to save the archival space. Advective tendencies need to be determined at the model's integration interval then averaged in time, to retain high-frequency variability. Second, the initial values of temperature

fields at all the model grid locations are needed. Time integration of the MLT tendency components with the temperature initial values will give T_1 and T_2 of Fig. 3, which are used to determine ΔT and Ent_volume according to Eqs. (8) and (13). It is worth noting that the MLT budget derived using the fields averaged over τ is not necessarily the same as the one from the model fields given at the model's original time stepping. For example, the mixed layer depth will be different between the two cases. At any averaging interval, however, our analysis will close the MLT budget. In this sense, our analysis is *exact*.

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